

LAB EXPERIMENT REPORT

Experiment No: 3

Date: 23/09/2026

Topic: Survey Report on Emerging Trends in Low-Code Application Development

Application: Event Registration & Participant Management

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1. AIM

To conduct a systematic survey on emerging trends in Low-Code Application Development (LCAD) platforms, evaluating their architectural paradigms, efficiency gains, and limitations compared to traditional software development models, with a specific focus on implementing an enterprise-grade **Event Registration & Participant Management System**.

2. TRADITIONAL VS. EMERGING TRENDS IN LOW-CODE APPLICATION DEVELOPMENT

Traditional application development relies on manual full-stack hand-coding, complex multi-tier architectures, and lengthy DevOps pipelines. In contrast, emerging Low-Code Application Development (LCAD) leverages visual drag-and-drop interfaces, modular component blocks, and automated workflow engine triggers.

Dimension	Traditional Development	Low-Code Development
Development Speed	Slow (Months). Bespoke hand-written code for frontends, backends, and schemas.	Rapid (Days/Weeks). Pre-built UI blocks, drag-and-drop forms, visual workflows.
Skill Level	High technical expertise (Java, React, SQL, Python, DevOps).	Citizen developers, business analysts, and accelerated pro-developers.
Integrations	Custom REST/GraphQL APIs and manual DB connections.	Pre-packaged API connectors (Stripe, SendGrid, CRM tools).
Maintenance	High effort; manual refactoring, patching, and framework migration.	Low-to-medium; cloud vendors manage core platform updates.
Event App Context	Bespoke form validation logic & custom QR scanner integration.	Drag-and-drop guest forms & automated ticket email triggers.

3. BENEFITS AND CHALLENGES IN LOW CODE

Key Benefits:

- Fast Deployment:** Event registration forms created and launched in hours.
- Cost Reduction:** Lower engineering overhead and shorter development cycles.
- Multi-Platform:** Native web and mobile experiences from a single build.
- Automations:** Instant payment confirmations, PDF badge generation, and check-in updates.

Key Challenges:

- Vendor Lock-in:** High friction and cost when migrating away from proprietary frameworks.
- Customization Limits:** Rigid UI templates for complex bespoke event interactions.
- Governance Risks:** Shadow IT risks regarding data security and GDPR compliance.
- Scalability Limits:** High surge loads require upgraded platform subscription tiers.

4. COMPARISON OF EXISTING PLATFORMS

A systematic comparison was conducted across five industry-standard low-code platforms to assess their viability for building an **Event Registration & Participant Management System**.

Platform	Target Audience	Strengths for Event Apps	Limitations	Deployment
OutSystems	Enterprise IT / Pro Devs	Offline check-in support, enterprise-grade data security.	High learning curve and expensive licensing.	Cloud / On-Premise
Mendix	Enterprise Business & IT	Native mobile QR badge scanning & collaborative model.	Complex pricing based on active user metrics.	Multi-Cloud / Hybrid
Power Apps	Citizen Developers / O365	Direct integration with Teams, SharePoint, and Outlook.	Higher cost model for external guest registrations.	Microsoft Azure
Appian	BPM & Workflow Teams	Strong process logic for VIP approval & speaker reviews.	Overkill for simple guest registration forms.	Cloud / On-Premise
Retool	Internal Tool Engineers	Rapid admin dashboards for real-time check-in tracking.	Not suited for public-facing guest marketing pages.	Self-Hosted / Cloud

5. RESULT & CONCLUSION

The survey highlights that low-code application development platforms reduce standard software development timelines by 60% to 70% compared to traditional manual coding methods.

For an **Event Registration & Participant Management System**, low-code platforms provide an ideal framework for handling form submissions, automated email confirmations, payment gateway integrations, and real-time check-in tracking. While bespoke full-stack programming remains necessary for hyper-custom consumer designs, low-code delivers the ideal balance of speed, operational agility, and maintainability for enterprise event management workflows.

Faculty Signature & Date

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